

Jahidul Islam

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Career Summary

I am a masters student in Computer Science with a focus on deep learning to sensor-driven and multimodal artificial intelligence. I have experience with the use of TensorFlow in time-series analysis. I am in the process of making a sign language recognition system using the tactile glove that combines pressure sensors, inertial measurement units, and the OptiTrack motion capture. I possess good skills in Python and C++, and have experience in embedded sensing technology, networking, and cybersecurity.

Education

The University of Aizu

Graduate School of Computer Science and Engineering

Fukushima, Japan

October 2024 – Present

- Research Assistant (Professor JING Lei's lab)
- Major in Computer Science and Engineering
- CGPA: 3.90/4.00 (Final Semester)
- Thesis: Sign Language Recognition Using a Data Glove Equipped with Pressure Sensors, IMU, and OptiTrack Motion Tracking.
- Relevant Coursework: Computer Communications and Networking, Performance Evaluation of Network Systems, Introduction to Deep Learning Acceleration, Image Recognition and Understanding, Information Security, and Machine Learning.

American International University–Bangladesh (AIUB)

Bachelor of Science in Computer Science and Engineering

Dhaka, Bangladesh

May 2017 – Mar 2023

- CGPA: 3.33/4.00
- Thesis: Phishing Attack Detecting System
- Relevant Coursework: Computer Networks, Advanced Computer Networks, Artificial Intelligence and Expert System, Network Security

Dhaka College

Higher Secondary School Certificate (HSC)

Dhaka, Bangladesh

Jul 2013 – Aug 2015

- Board: Dhaka
- Group: Science
- GPA: 5.00

Homna Government High School

Secondary School Certificate (SSC)

Cumilla, Bangladesh

Jan 2011 – May 2013

- Board: Cumilla
- Group: Science
- GPA: 5.00

Research Interests

- Deep Learning and Machine Learning
- Time-Series Learning for Wearable and Tactile Sensors
- Multimodal AI (Tactile, IMU, Motion/Vision Fusion)
- Efficient / Edge AI for Embedded Systems
- AI for Healthcare (Medical Imaging and Clinical Data)
- Security, Networking, and IoT Systems

Publications

- **Islam, J.**, & Alif, M. D. A. (2025). *An efficient deep learning model for static ASL gesture recognition in resource-constrained environments*. In *Proceedings of the 2025 International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN)* (pp. 1–6). IEEE. <https://doi.org/10.1109/QPAIN66474.2025.11172129>
- Watanabe, A., **Islam, J.**, & Jing, L. (2025). *Enhanced nondestructive prediction of peach sugar content using hyperspectral imaging and convolutional neural networks*. In *Proceedings of the 2025 IEEE 14th Global Conference on Consumer Electronics (GCCE)* (pp. 1–6). IEEE. <https://doi.org/10.1109/GCCE65946.2025.11275303>
- Watanabe, R., **Islam, J.**, Zhu, X., Kaneko, E., Iseki, K., & Jing, L. (2025). *Chest compression skill evaluation system using pose estimation and web-based application*. *Applied Sciences*, 15(15), 8252. <https://doi.org/10.3390/app15158252>

Technical Skills

- **Programming Languages:** Python, C++, Java, C#
- **Deep Learning / ML:** TensorFlow, model training & evaluation, time-series classification
- **Sensor & Embedded Systems:** ESP32, tactile/pressure sensors, IMU sensors, Flex sensor, sensor data acquisition, sensor readout circuits
- **Measurement & Testing:** IMADA force/pressure testing machine (sensor calibration/controlled force experiments)
- **Motion Capture / Vision:** OptiTrack (Motive), pose estimation
- **Prototyping:** 3D printing, Bambu Studio (slicing/print preparation)
- **Design Tools:** PE-DESIGN 11 (tactile sensor layout/pattern drawing)
- **Fabrication Tools:** Brother embroidery machine (conductive-thread tactile sensor fabrication)
- **Web Development:** HTML, CSS, JavaScript, PHP, MySQL
- **Database Systems:** SQL, Oracle
- **Cybersecurity Tools:** Kali Linux, Oracle VirtualBox, Wireshark
- **Networking:** Network Troubleshooting, Firewall Configuration, Penetration Testing

Work Experience

The University of Aizu
Teaching Assistant

Fukushima, Japan
September 2024 – Present

- **Introduction to Computer Networks:** Lab assistant, assisted with and demonstrated network setup to students and helped with assignments.
- **Operating Systems:** Guided students to learn the concept of process management, memory handling and aided practical tasks.
- **Data Structures and Algorithms:** Helped students with their coding exercises, taught basic algorithms as well as problem-solving.
- **Artificial Intelligence :** Helped students with coding exercises, and was responsible for evaluating quizzes and grading reports.

The University of Aizu
Research Student

Fukushima, Japan
April 2024 – September 2024

Graduate School of Computer Science and Engineering

- Research student at the Faculty of Computer Science and Engineering, University of Aizu, engaged in research activities and gaining practical knowledge in advanced Computer Science methodologies.

American International University–Bangladesh (AIUB)
Teaching Assistant (Internship)

Dhaka, Bangladesh
January 2022 – April 2022

- Assisted students in Object-Oriented Programming (Java) labs.
- Resolved software and database issues.
- Reported on lab equipment for potential improvements.

Thesis & Projects

Master Thesis - Hysteresis Analysis of Piezoresistive Pressure Sensors for Improved Sign Language Recognition Using Deep Learning

This research explores the impact of hysteresis-based feature extraction on dynamic sign language recognition and compares deep learning model accuracy using hysteresis-derived data versus raw sensor data.

Undergraduate Thesis - Phishing Attack Detecting System

This research aims to enhance phishing attack detection for secure browsing by addressing various attack categories, leveraging a reliable prototype for future development.

Projects

- **Globeride Project:** Data Glove Development (Designed a data glove for Globeride, tested it using IMADA, calibrated the sensors, and prepared it for real-time raw sensor data collection).
- **Database:** Superstore Management (Optimizes buying, billing, and secure data storage for efficient operations).
- **Computer Graphics:** Traffic Signal in the City (Its primary function is the management of traffic signals within a specific area).

- **C#:** Study Assist Project (How to effectively manage the education system in an institution).
- **Object-Oriented Programming (Java):** Bookshop Management (Join the library to explore books and authors more comprehensively).
- **Object-Oriented Programming (Java):** Restaurant Management (In restaurant management, explore daily menus, and stay updated on food offers).

All the projects are available on GitHub: <https://github.com/jahidaiub06>

References

Dr. Jing Lei

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The University of Aizu, Japan

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